

Software Requirements Specification

Project 27 | Project Management System | Version 1.0 | 28 September 2026

Prepared for the Software Engineering mini-project. Team and approval details may be completed by the submitting group.

1 Introduction

1.1 Purpose

Defines the externally observable behavior and measurable quality goals for a web-based Project Management System. It is the baseline for the design and test plan.

1.2 Scope

The system manages projects, members, tasks, a Kanban workflow, time estimates and logged effort, progress summaries, and a CSV export. Version 1 excludes billing, file attachments, external calendar integration, and automated scheduling.

1.3 Audience and definitions

Audience: project stakeholders, developers, testers and evaluators. Owner controls a project; Manager coordinates tasks; Member contributes to assigned tasks. FR means functional requirement; NFR means nonfunctional requirement; SR means security requirement.

2 Overall description

A browser communicates with a server-side application and relational database. A user may belong to multiple projects; all queries are scoped to the selected project. Core roles are Owner, Manager and Member.

2.1 Product functions

Account access, project membership, task lifecycle, Kanban board, estimates, time logs, comments, search, activity history, dashboard, notifications, export and archive.

2.2 Environment and constraints

Desktop browser with network access; Django web application and relational database are the proposed implementation. Dates use stored UTC timestamps and a displayed user time zone. A project is archived rather than deleted to preserve history.

3 External interface requirements

Web UI: login, project list, project settings, board, task detail and dashboard. Server API: JSON over HTTPS; CSV download for export. Database: Users, Projects, Memberships, Tasks, TimeEntries, Comments, Events and Notifications. No special hardware is required.

4 System features and functional requirements

Each statement describes one testable outcome. A request without sufficient project permission is denied server-side.

4.1 Account registration (FR-01)

The system shall allow a new user to register with a unique email address and password; it shall reject an email already registered.

Acceptance: A new email creates one account; a duplicate returns a conflict without a second account.

4.2 Authentication (FR-02)

The system shall allow registered users to sign in and sign out using email and password.

Acceptance: Correct credentials open a session; invalid credentials do not.

4.3 Project creation (FR-03)

The system shall allow a signed-in user to create a project with a name, description, start date and optional target date; the creator becomes Owner.

Acceptance: A valid submission creates one project owned by its creator.

4.4 Membership (FR-04)

The Owner shall invite an existing user by email and assign Owner, Manager, or Member role; only an Owner shall change membership roles.

Acceptance: Invited user receives access; a Member role-change attempt is denied.

4.5 Task creation (FR-05)

An Owner or Manager shall create a task with title, description, priority, assignee, estimate in hours and due date in a project to which they belong.

Acceptance: Valid data creates a task with a unique ID and Backlog status.

4.6 Task editing (FR-06)

An Owner or Manager shall edit task fields; a Member shall edit only tasks assigned to them, excluding assignee and estimate.

Acceptance: Allowed edits persist; forbidden field changes are rejected.

4.7 Kanban transition (FR-07)

A permitted project member shall move a task among Backlog, To Do, In Progress, Review and Done; every transition shall be recorded.

Acceptance: Board and task detail reflect new state with actor and time in history.

4.8 Board filtering (FR-08)

The system shall filter the project board by assignee, priority and due status, separately or together.

Acceptance: Only tasks matching every selected filter appear.

4.9 Estimation (FR-09)

An Owner or Manager shall set or change a nonnegative estimate in 0.5-hour increments for a task.

Acceptance: Negative or nonincrement values are rejected; valid values persist.

4.10 Time logging (FR-10)

A project member shall log positive time in 0.25-hour increments against a task, with date and note, and shall edit only their own entries.

Acceptance: Actual hours equals sum of retained entries and updates after edits.

4.11 Progress visibility (FR-11)

The project dashboard shall show counts by status, overdue task count, estimated hours and actual hours based on current project data.

Acceptance: Displayed totals match a query of the same project records.

4.12 Task comments (FR-12)

A project member shall add a nonempty comment to a task and see comments in chronological order.

Acceptance: Author, timestamp and text appear on task detail.

4.13 Activity history (FR-13)

The system shall record project membership, task creation, edits and status changes with actor, action, entity and timestamp.

Acceptance: History entries are available to project members in time order.

4.14 Search (FR-14)

A project member shall search tasks in their projects by title or task ID, with project-scoped results.

Acceptance: Matching authorized tasks appear; other projects are excluded.

4.15 Export (FR-15)

An Owner or Manager shall download a CSV of their project tasks containing ID, title, status, assignee, estimate, actual hours and due date.

Acceptance: CSV row count and values match project task records.

4.16 Archive project (FR-16)

An Owner shall archive a project; archived projects shall be read-only for members and absent from active boards.

Acceptance: Reads remain available; attempts to modify return a conflict.

4.17 Notifications (FR-17)

The system shall create an in-app notification when a member is assigned a task or invited to a project.

Acceptance: The target user sees exactly one notification for each successful event.

5 Nonfunctional requirements

ID	Quality	Measurable requirement
NFR-01	Performance	For a project with 500 tasks and 20 members, 95% of board and task-list requests shall complete within 2 seconds in a local test environment under 20 concurrent users.
NFR-02	Capacity	A project shall support at least 500 tasks, 20 members, and 2,000 activity events without loss or truncation.
NFR-03	Reliability	A committed task update shall remain available after application restart; database transactions shall roll back completely on an injected failure.
NFR-04	Usability	At 1280 × 720 desktop resolution, a new user shall be able to create a project and task in at most 10 interface actions each.
NFR-05	Accessibility	Core create, search and board actions shall work with keyboard alone; controls shall have visible focus and programmatic labels.
NFR-06	Compatibility	The web interface shall complete core flows on the latest stable Chrome and Firefox available at test execution.

5.1 Security

5.1.1 Security objectives

SO-01: Prevent users from viewing or changing projects they do not belong to. SO-02: Protect credentials and sessions from disclosure or misuse. SO-03: Preserve trustworthy task change history.

5.1.2 Security requirements

ID	Requirement
SR-01	All project and task reads/writes shall check authenticated project membership on the server; unauthorized access returns 403 or 404 without revealing records.
SR-02	Passwords shall be stored with a salted adaptive hash; plaintext passwords shall not appear in database fields or application logs.
SR-03	State-changing browser requests shall use CSRF protection, and session cookies shall be HttpOnly and SameSite=Lax or stricter.
SR-04	User supplied text shall be encoded on output and validated for maximum length to prevent stored script execution.
SR-05	After five failed sign-in attempts for an account within 15 minutes, further attempts shall be temporarily blocked for 15 minutes and an event recorded.

6 Quality attributes and acceptance

Release acceptance requires all high-priority functional cases and security cases to pass, zero open critical defects, and verified traceability for each SRS ID. Performance thresholds use the stated fixture and concurrent load.

7 System models and diagrams

7.1 Project planning use cases

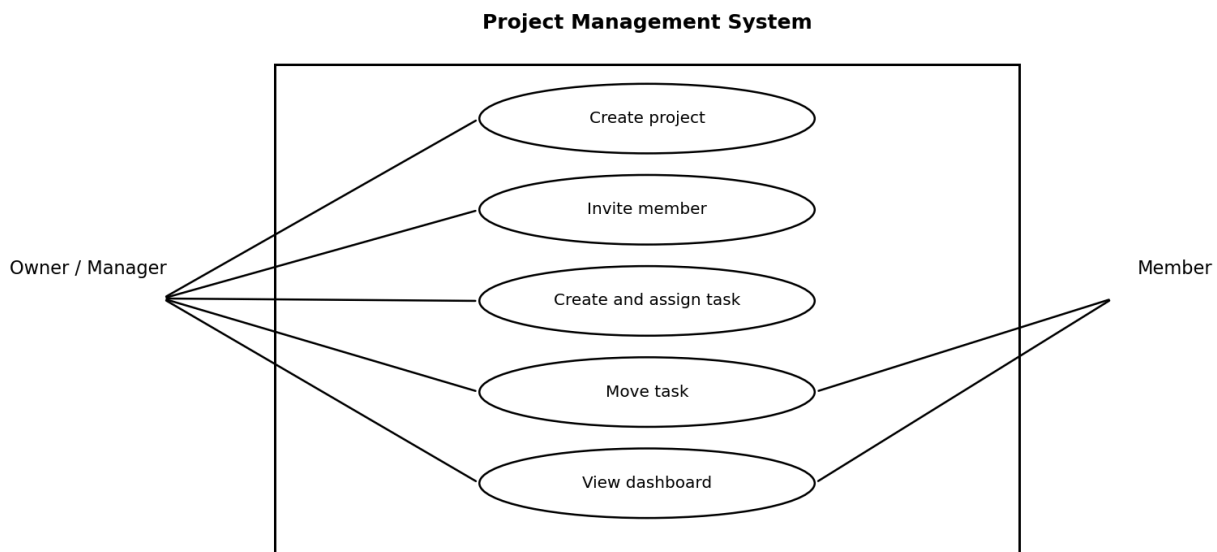


Figure 1. Planning, task creation and board visibility use cases.

7.2 Collaboration and reporting use cases

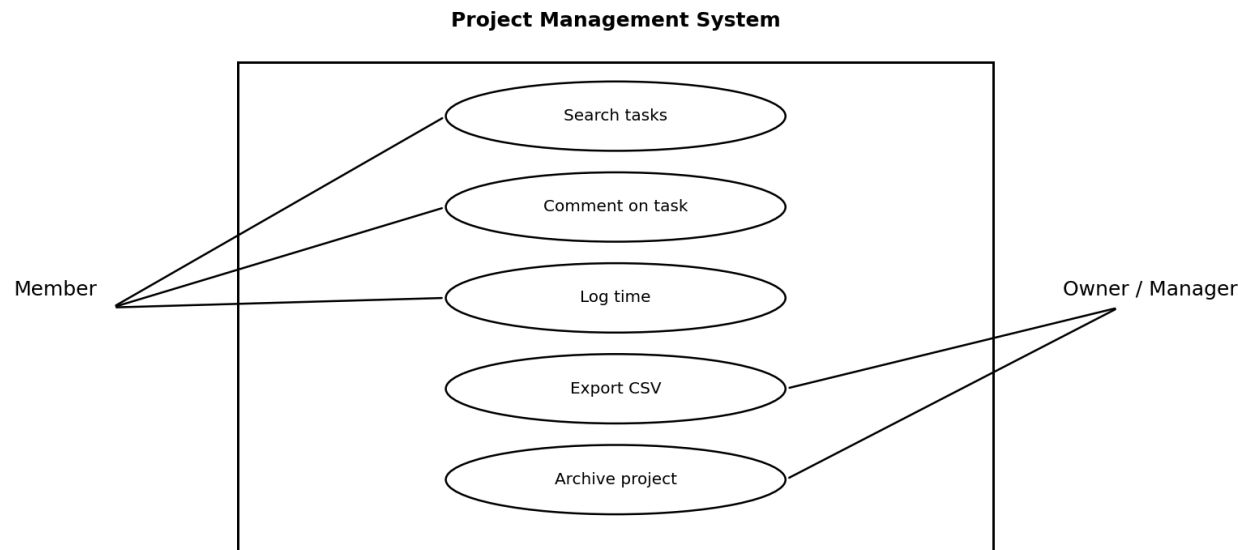


Figure 2. Task collaboration and reporting use cases.

8 Requirements traceability matrix

Requirement	Design location	Validation
FR-01	SAD 3-4	TC-01
FR-02	SAD 3-4	TC-02
FR-03	SAD 3-4	TC-03
FR-04	SAD 3-4	TC-04
FR-05	SAD 3-4	TC-05
FR-06	SAD 3-4	TC-06
FR-07	SAD 3-4	TC-07
FR-08	SAD 3-4	TC-08
FR-09	SAD 3-4	TC-09
FR-10	SAD 3-4	TC-10
FR-11	SAD 3-4	TC-10
FR-12	SAD 3-4	TC-12
FR-13	SAD 3-4	TC-07
FR-14	SAD 3-4	TC-12
FR-15	SAD 3-4	TC-13
FR-16	SAD 3-4	TC-13
FR-17	SAD 3-4	TC-14
NFR-01	SAD 3-4	TC-16
NFR-02	SAD 3-4	TC-16
NFR-03	SAD 3-4	TC-17
NFR-04	SAD 3-4	TC-18
NFR-05	SAD 3-4	TC-18
NFR-06	SAD 3-4	TC-18
SR-01	SAD 3.9	TC-04
SR-02	SAD 3.9	TC-15
SR-03	SAD 3.9	TC-15
SR-04	SAD 3.9	TC-15
SR-05	SAD 3.9	TC-15